# Chemistry Playlist

Pre-Test:

Atomic Numbers - https://www.youtube.com/watch?v=7qrZti505B4
Balancing part 1: https://www.youtube.com/watch?v=UGf60kq_ZDI
Balancing part 2: https://www.youtube.com/watch?v=kjiB8Amic9I

Ch 2
2.1 Physical vs Chemical changes: https://www.youtube.com/watch?v=ARB2fZHutas
2.2 Accuracy Precision : https://www.youtube.com/watch?v=PMv69Gne_Bw
2.2a Unit Conversions - https://www.youtube.com/watch?v=lbEn0G-7S4M
2.2a1 – Powers of Ten - https://www.youtube.com/watch?v=44cv416bKP4
2.2b – Scale of units - https://www.youtube.com/watch?v=-aYat9357mE
2.2c – Scientific notation TI-84 - https://www.youtube.com/watch?v=Hr6APFR1qpo
2.3 Sig Figs: https://www.youtube.com/watch?v=LEBSq_oC9Jk
2.4 Density: https://www.youtube.com/watch?v=REtBibhIqfo
Metric SI song 😊 https://www.youtube.com/watch?v=Fcl4gcGN6kk

Lab – vernier caliper - https://www.youtube.com/watch?v=PfFUmUeLQa4
Lab – Vernier caliper part 2 - https://www.youtube.com/watch?v=UiPfVdymjAE
Lab – buret part 1 – https://www.youtube.com/watch?v=spxFG_xQjhE
Lab – buret part 2 - https://www.youtube.com/watch?v=DOqP5Zd4WMY

Ch 3
3.2a History of the atom part 1: https://www.youtube.com/watch?v=sG6QoLxwIw4
3.2b History of the atom part 2: https://www.youtube.com/watch?v=GhAn8xZQ-d8
3.3a Atomic number, pne, isotopes: https://www.youtube.com/watch?v=7qrZti505B4
3.3b Average atomic mass (isotopes): https://www.youtube.com/watch?v=G6A4UQX7QRw
3.3c Interconverting masses, moles and number particles :
https://www.youtube.com/watch?v=tBbCX6dQZPo
3.3d – Mass spectrometer:
https://www.youtube.com/watch?v=mBT73Pesiog&list=PLllVwaZQkS2op2kDuFifhStNsS49LAxkZ&index=10&t=0s
3.3e – Avogadro's Number - https://www.youtube.com/watch?v=TSQlM72_MiU
3.3f – Brownian motion - https://www.youtube.com/watch?v=ZNzoTGv_XiQ
3.4a Nuclear decay – general – https://www.youtube.com/watch?v=c9WfZJYUWv0
3.4b Nuclear decay – balanced equations – https://www.youtube.com/watch?v=CaYoDxWxww8
3.4c Nuclear power - https://www.youtube.com/watch?v=rcOFV4y5z8c
3.4d Nuclear bombs: - https://www.youtube.com/watch?v=fYuVzbIu_8o
3.5 Helium reserve - https://www.youtube.com/watch?v=mOy8Xjaa_o8
3.6 Soviet nuclear weapons used - https://www.youtube.com/watch?v=QlVmo_jvBQE

Mole Day 😊 - https://www.youtube.com/watch?v=CPR9eZBpCW4

Ch 4 – Electron configuration and theory
4.1 Emission Spectra - https://www.youtube.com/watch?v=1uPyq63aRvg
4.1a. Light - problems c = λν- https://www.youtube.com/watch?v=goJl54Y1hio
4.1b. Quantized and Energy – Photons : https://www.youtube.com/watch?v=VLc20TF9jZw
4.2a - Heisenberg - https://www.youtube.com/watch?v=t8mMN2X5_Vw
4.2b – Schrodinger equation - https://www.youtube.com/watch?v=O6g-7rUgrdg

4.2c Quantum numbers - https://www.youtube.com/watch?v=Aoi4j8es4gQ
4.3a Electron Configuration part 1 : https://www.youtube.com/watch?v=8n2CXyEZIlE
4.3b Electron Configuration part 2 : https://www.youtube.com/watch?v=diCGRJskeDA
4.4 Rydberg equation : Preview of APChem https://www.youtube.com/watch?v=au2HCVn9IJI
Schrodinger's Cat - https://www.youtube.com/watch?v=UjaAxUO6-Uw&t=61s
4.5 Speed of light - https://www.youtube.com/watch?v=pTn6Ewhb27k
4.6 Quantum – multiverse - https://www.youtube.com/watch?v=h6OoaNPSZeM
4.7 – Big Bang evidence - https://www.youtube.com/watch?v=zmrbjp-GDfk

Mysteries of the Universe : https://www.youtube.com/watch?v=Jn7VcOU3x2g
Watch 00:00 – 60:00 and 1:20:00 – 1:50:00 (or all of it ☺)

Neil DeGrasse Tyson – Aliens - https://www.youtube.com/watch?v=Dkjkh3OrjeA

Ch 5 – Periodicity and electrons

5.1 – History of Periodic Table – https://www.youtube.com/watch?v=fPnwBITSmgU&t=20s
5.2 – Groups and Group Properties – https://www.youtube.com/watch?v=fLSfgNxoVGk&t=1s
5.3 – 4 Trends Atomic radius, EN, IE, Ionic radius ☺ -
5.3a - https://www.youtube.com/watch?v=hePb00CqvP0
5.3b - https://www.youtube.com/watch?v=XK-WTYncldA
5.3c -ionic radius - https://www.youtube.com/watch?v=lzKJWwTDU34
5.3d ☺ https://www.youtube.com/watch?v=qc02d3G_wb4
5.e3 – Ionic radius trick - …. https://www.youtube.com/watch?v=9PzMiy4Mt-Q
5.5 – PES diagram - https://www.youtube.com/watch?v=gyKD6QULa0A

Ch 6 – Bonding

6.1 – Types of bonds – https://www.youtube.com/watch?v=PoQjsnQmxok
6.2 - Covalent – bond energy - https://www.youtube.com/watch?v=Mo4Vfqt5v2A
6.2- Covalent – Lewis intro - https://www.youtube.com/watch?v=cIuXl7o6mAw
6.2 – Covalent – Lewis part 1 – https://www.youtube.com/watch?v=qCOkHYYc4mE
6.2 – Covalent – Lewis part 2 - https://www.youtube.com/watch?v=zEQeVvdjkBc
6.3 – Ionic vs covalent - https://www.youtube.com/watch?v=7DjsD7Hcd9U
6.3 – Ionic bonding (lattice energy) - https://www.youtube.com/watch?v=hiyTfhjeF_U
6.4 – Metallic bonding - https://www.youtube.com/watch?v=eVv3TpaQ2-A
6.4 – metal properties - https://www.youtube.com/watch?v=vOuFTuvf4qk
6.5b - VSEPR part 1 : https://www.youtube.com/watch?v=nxebQZUVvTg
6.5c – VSEPR part 2- https://www.youtube.com/watch?v=Moj85zwdULg
6.5d - VSEPR part 3- https://www.youtube.com/watch?v=Q9-JjyAEqnU
6.5e – polar nonpolar - https://www.youtube.com/watch?v=OHFGXfWB_r4
Also https://www.youtube.com/watch?v=72CQe-_PJU4

6.5d IMF - https://www.youtube.com/watch?v=08kGgrqaZXA
Also https://www.youtube.com/watch?v=-QqTwJzi7Wo
Also https://www.youtube.com/watch?v=S8QsLUO_tgQ

Lewis Structures (MM) part x : https://www.youtube.com/watch?v=XbSW2x35OeU
6.2 – Covalent – Lewis part 0 - https://www.youtube.com/watch?v=ulyopnxjAZ8
6.6 – Gecko – LDF - https://www.youtube.com/watch?v=gzm7yD-JuyM
6.6 – Gecko 2 – LDF -- https://www.youtube.com/watch?v=TlyvS1ckDZM
6.6 – Gecko 3 – LDF -- https://www.youtube.com/watch?v=IkV1zMh_GQk
6.6 – Snowflake bonding - https://www.youtube.com/watch?v=ao2Jfm35XeE
6.6 – surface tension - https://www.youtube.com/watch?v=ynk4vJa-VaQ

Ch 7 Compounds
7-1a: Polyatomic ions: https://www.youtube.com/watch?v=p9iQ5Qn42DM
7-1b: Naming ionic – part 1: https://www.youtube.com/watch?v=G3PNjgeKhoM
7-1c: Naming ionic – part 2: https://www.youtube.com/watch?v=KKAh8oamp94
7-1d: covalent/ionic mixed: https://www.youtube.com/watch?v=0CT0MoBmT3w
7-2a: Oxidation State part 1 - https://www.youtube.com/watch?v=w0RfMRDy34w
7-2b: Oxidation State part 2 - https://www.youtube.com/watch?v=-a2ckxhfDjQ
7-3a: Percent Composition part 1- https://www.youtube.com/watch?v=lywmGCfIUIA
7-3b: Percent Composition part 2 - https://www.youtube.com/watch?v=L8ImtVPQDDQ
7-3c: HOOD-Program 20 -
https://www.youtube.com/watch?v=pdavIrUnc0s&feature=youtu.be

7-4a: Empirical Formula part 1 - https://www.youtube.com/watch?v=AFqwtY7m2PI
7-4b: Empirical Formula part 2- https://www.youtube.com/watch?v=wnRaBWvhYKY
7-4c: HOOD-Program 21 -
https://www.youtube.com/watch?v=WuUDjSSZ0Rk&feature=youtu.be

Lab – Hydration number - https://www.youtube.com/watch?v=UwJ75hfQiXY

Ch 8 Chemical Reactions
8-1: Balancing Reactions - https://www.youtube.com/watch?v=eNsVaUCzvLA
Balancing part 1: https://www.youtube.com/watch?v=UGf60kq_ZDI
Balancing part 2: https://www.youtube.com/watch?v=kjiB8Amic9I
8-2: 5 types - https://www.youtube.com/watch?v=aMU1RaRulSo
8-2 – Double replacement - https://www.youtube.com/watch?v=BV2dmSwmGpQ
8-3 Activity Series - https://www.youtube.com/watch?v=AU2c-9D-Cj8
8-3b. - https://www.youtube.com/watch?v=A94EJfPaCf8
8-3c: - https://www.youtube.com/watch?v=Vl950rSnYos

Catalytic converter 1 - https://www.youtube.com/watch?v=HADOcrcMikA
Catalytic converter 2 - https://www.youtube.com/watch?v=R9G2erJMmmQ
Catalytic converter 3 - https://www.youtube.com/watch?v=rmtFp-SV0tY
Catalytic converter 4 - https://www.youtube.com/watch?v=W6dIsC_eGBI
Haber Process - https://www.youtube.com/watch?v=EvknN89JoWo
Lead toxicity - https://www.youtube.com/watch?v=IV3dnLzthDA
Biofuels - https://www.youtube.com/watch?v=OpEB6hCpIGM
Anode Rod – oxidation - https://www.youtube.com/watch?v=2IUNIUZz4Os
Ozone layer reforming - https://www.youtube.com/watch?v=CaLOiGEDPJQ

Ch 9 Stoichiometry – FUN!
9-1: mole ratios-
9-2a: mole-mole problems- Program 24
https://www.youtube.com/watch?v=G9z89uFtGRY&feature=youtu.be
9-2b: mass-mass problems (gram-gram). – Program 25
https://www.youtube.com/watch?v=JZALxSIYFZ8

//old https://www.youtube.com/watch?v=V6gbMFxEHAg&feature=youtu.be

9-3: Limiting/excess/%Yield problems ---
9.3a – intro - https://www.youtube.com/watch?v=nNhzosyqNVI
9.3b – 2 more problems - https://www.youtube.com/watch?v=q0M1ZTSYac0

Ch 10 Heat

10.1a – Heat - Q=mcT - https://www.youtube.com/watch?v=eIBVimdYnDI
10.1b – Heat part 2 - https://www.youtube.com/watch?v=yhNHJ7WdT8A
10.2 – phase diagram - https://www.youtube.com/watch?v=oc0ypeDELb0

16.1 – Enthalpy intro endo vs exo - https://www.youtube.com/watch?v=KCQALFuAZOc
16.1a – Enthalpy Heat of Reaction part 1 - https://www.youtube.com/watch?v=PJweE3F_mPY
16.1b – Heat of Reaction part 2 - https://www.youtube.com/watch?v=61fKkCwWu4s
Lab – Heat of Reaction : https://www.youtube.com/watch?v=pjFV-A7wVJI

Ch 11 – Gases
11.1a – Gas Pressure Units - https://www.youtube.com/watch?v=YQ0ib6XlSyk
11.1b – Reading Barometer – https://www.youtube.com/watch?v=OeCvgRbbY1Q
11.1c – Dalton' Law Partial Pressures - https://www.youtube.com/watch?v=JbqtqCunYzA
11.2a – Gas Laws part 1 KMT – https://www.youtube.com/watch?v=o3f_VJ87Df0
11.2b – Gas Laws part 2 –
11.2c – Gas Law problems - https://www.youtube.com/watch?v=xL9KclCmC1k
11.2d – Gas law problems part 2 - https://www.y outube.com/watch?v=o8dMAWfM1Ag

11.2e – Avogadro's Number - https://www.youtube.com/watch?v=TSQlM72_MiU
11.3a – Ideal Gas law intro – R - https://www.youtube.com/watch?v=kBgzzwCTddc
11.3b – Idea gas law part 2 - https://www.youtube.com/watch?v=WhP6zJbSxec
11.3b – Ideal Gas law problems – https://www.youtube.com/watch?v=yn7kmeD47Zc
11.3c – Ideal Gas Law more problems – https://www.youtube.com/watch?v=TqLlfHBFY08
11.4 - Graham's Law of Effusion/Diffusion - https://www.youtube.com/watch?v=NniisV69jYk
11.4 – Graham's Law of Diffusion – https://www.youtube.com/watch?v=LWZYh-tSYgw
11.4c – SF6 gas 😊 https://www.youtube.com/watch?v=u19QfJWI1oQ
11.4d – another - https://www.youtube.com/watch?v=U4ixGhtBPp0
11.4e – instruments played… - https://www.youtube.com/watch?v=GIamVw6Va3o

Lab – Gas production - https://www.youtube.com/watch?v=QKMJpTG_YIs

Ch 12 – Solutions

12.3a – Molarity – https://www.youtube.com/watch?v=yb4FW6E1HKE
12.3b – Molarity – dilution – https://www.youtube.com/watch?v=v6dnEp58mVk
12.3c – molality - https://www.youtube.com/watch?v=Dc-seqQg_as

Ch 13 –
13-1a – Dissociation ionic - https://www.youtube.com/watch?v=6APgDoZ_qJY
13-1b – Precipitates – Ionic, Net Ionic and Spectator Ions - https://www.youtube.com/watch?v=MJy5CNzd6sY

13-2 – Colligative properties (BP, FP, osmotic pressure)
13-2a – https://www.youtube.com/watch?v=hOhgXRukRDg
13-2b – https://www.youtube.com/watch?v=c8dDLe37ONg
13-2c - https://www.youtube.com/watch?v=wzAToPL8Src

14 Acid base
14.1 – Acid Base Properties - https://www.youtube.com/watch?v=b-n2_QxX8po
14.1a- Naming acids 1 - https://www.youtube.com/watch?v=qSbNlne-5P0
14.1b – Naming acids 2 - https://www.youtube.com/watch?v=Dqdo0gT5yG0
14.2 – Neutralization - https://www.youtube.com/watch?v=0cwh40YAtcg
14.3 – Acid Base Definitions - https://www.youtube.com/watch?v=EyBkPwsRY2E

Ch 15 – pH and Titrations

15.1 Strong-weak acids- https://www.youtube.com/watch?v=_gYBbzkqrmE
15.1a – Strong acids (trick) - https://www.youtube.com/watch?v=jPilrd7b1B8
15.2 Acid base-pH-pOH : https://www.youtube.com/watch?v=NNXvokAcSuE
15.2a – pH pOH problems: https://www.youtube.com/watch?v=usk4Sl2Jw9M
15.3 – Weak acid Ka : https://www.youtube.com/watch?v=7C_HsfB_6PQ
15.4 – Titration problems : https://www.youtube.com/watch?v=70vlWskHfc4

Lab – titration - https://www.youtube.com/watch?v=Ecy2FafmKI4

10.4 Another phase diagram - https://www.youtube.com/watch?v=QrHlwgmMTq4

Electron microscopes - https://www.youtube.com/watch?v=9DnnxvS6BBQ

# ChemAP Playlist

Ch 01-02 – Intro
AP Chem Playlist:

Ch 1 – Sig figs:

1.5 Sig Figs: https://www.youtube.com/watch?v=LEBSq_oC9Jk
1.8 Density: https://www.youtube.com/watch?v=REtBibhIqfo
1.9a Unit Conversions - https://www.youtube.com/watch?v=lbEn0G-7S4M
1.9b Units – scale - https://www.youtube.com/watch?v=-aYat9357mE

Ch 2 – History of the Atom and Naming Compounds:
2.3 History of the atom part 1: https://www.youtube.com/watch?v=sG6QoLxwIw4
2.4 History of the atom part 2: https://www.youtube.com/watch?v=GhAn8xZQ-d8
2.5 - Atomic Numbers - https://www.youtube.com/watch?v=7qrZti505B4
2.5a – Avogadro's Number - https://www.youtube.com/watch?v=TSQlM72_MiU
2.8a Naming ionic – part 1: https://www.youtube.com/watch?v=G3PNjgeKhoM
2.8b Naming ionic – part 2: https://www.youtube.com/watch?v=KKAh8oamp94
2.8c Naming covalent - https://www.youtube.com/watch?v=0CT0MoBmT3w
2.8d. Naming acids - https://www.youtube.com/watch?v=AaYQ_ZiE7AY
2.8e – Strong acids - https://www.youtube.com/watch?v=jPilrd7b1B8
2.8f – Strong acids https://www.youtube.com/watch?v=-M5xuzSJCjs

Ch 3 – Stoichiometry…

3.1a Average atomic mass (isotopes): https://www.youtube.com/watch?v=G6A4UQX7QRw
3.1b Mass Spectrometry : Bozeman Sci 09 - https://www.youtube.com/watch?v=mBT73Pesiog&list=PLllVwaZQkS2op2kDuFifhStNsS49LAxkZ&index=10&t=0s
3.2 Interconverting masses, moles and number particles : https://www.youtube.com/watch?v=tBbCX6dQZPo
3.2 The Mole : Bozeman Sci 03 : https://www.youtube.com/watch?v=Pft2CASl0M0&list=PLllVwaZQkS2op2kDuFifhStNsS49LAxkZ&index=3
3.4 – Percent Composition from Formulas : https://www.youtube.com/watch?v=L8ImtVPQDDQ
3.5 – Empirical and Molecular Formulas : https://www.youtube.com/watch?v=AFqwtY7m2PI
3.5a – Emp and Mol. Formula - https://www.youtube.com/watch?v=eHR9saDmeYk
3.9a – Limiting Reagent - https://www.youtube.com/watch?v=IvCPLCQ-YK0
3.9b - Introduction to Limiting/Excess Reactant - https://www.youtube.com/watch?v=nZOVR8EMwRU
3.9c - theoretical actual and percent yield practice problems - https://www.youtube.com/watch?v=mmsKDK9WXdE

3.5b Bozeman Sci 02 – Chemical Analysis :
https://www.youtube.com/watch?v=QcC4OsSxWYU&list=PLllVwaZQkS2op2kDuFifhStNsS49LAxkZ&index=2

Ch 4 Solutions:
4.1 Dilution - https://www.youtube.com/watch?v=v6dnEp58mVk
4.2a Ionic solutions - https://www.youtube.com/watch?v=7fHA17DOrBg
4.2b Ionic solutions (more) - https://www.youtube.com/watch?v=6APgDoZ_qJY
4.5 a – Ionic, Net Ionic and Spectator Ions - https://www.youtube.com/watch?v=MJy5CNzd6sY
4.5 b – Molecular, Ionic and Net Ionic Equations :
https://www.youtube.com/watch?v=dvupBubB-HQ

4.8 a – Acid base neutralization –
https://www.youtube.com/watch?v=HdmCagtasYg&list=PLllVwaZQkS2op2kDuFifhStNsS49LAxkZ&index=30
4.8 b – Basic titrations – https://www.youtube.com/watch?v=dLNsPqDGzms
4.9 a – Redox – oxidation numbers: https://www.youtube.com/watch?v=w0RfMRDy34w
4.9 b – Redox – oxidations numbers part 2:
https://www.youtube.com/watch?v=VXvtkwubQQg
4.9 c – Redox – reactions -- https://www.youtube.com/watch?v=RX6rh-eeflM
4.10a – Redox – Balance in Acidic solutions: https://www.youtube.com/watch?v=IZ1tKxsqV74
4.10b – Redox – another. https://www.youtube.com/watch?v=HGzy_pRyej8
4.10c – Redox – yet another: https://www.youtube.com/watch?v=N6ivvu6xlog
4.11 – Anode Rod - https://www.youtube.com/watch?v=2IUNIUZz4Os
Redox online lab – part 1. -- https://www.youtube.com/watch?v=th6UI2SKIXU
Redox online lab – part 2-- https://www.youtube.com/watch?v=TQlgiD8qOFU

Ch 5 Gases:
5.1 Gases - Pressures - https://www.youtube.com/watch?v=qv81QCGNnVo
5.1 Gases – reading barometer - https://www.youtube.com/watch?v=OeCvgRbbY1Q
5.2 Pressure, V, T relationships: https://www.youtube.com/watch?v=ir64EcRkf5Q
5.3 Idea gas law - https://www.youtube.com/watch?v=gmN2fRlQFp4
5.3a PVT problems - https://www.youtube.com/watch?v=xL9KclCmC1k
5.3b PV-nRT problems -- https://www.youtube.com/watch?v=TqLlfHBFY08
5.3c – R derived - https://www.youtube.com/watch?v=kBgzzwCTddc
5.5 Dalton's Law (go to 9:00 min) - https://www.youtube.com/watch?v=JbqtqCunYzA
5.7 Graham's Law Effusion/Diffusion. https://www.youtube.com/watch?v=J2LNU9hGsHc
5.7 Graham's Law another problem. https://www.youtube.com/watch?v=NniisV69jYk
5.8 Real gas (non-ideal ). https://www.youtube.com/watch?v=8zJrjEV9n8o&t=90s
5.9 Brownian motion - https://www.youtube.com/watch?v=ZNzoTGv_XiQ
Virtual lab – Mg and HCl react…. -- https://www.youtube.com/watch?v=QKMJpTG_YIs

Ch 6 – Heat:

6. 2 – Q=mcT - https://www.youtube.com/watch?v=eIBVimdYnDI
6.3a Hess' Law - https://www.youtube.com/watch?v=2q2u5sj4V00
6.3b Hess law 2 - https://www.youtube.com/watch?v=Z4ntAj7GcIo
6.3c Hess law 3 - https://www.youtube.com/watch?v=iETCSFit-zA
6.3d Hess Law 4 - https://www.youtube.com/watch?v=xkDkYX6eRbE
6.4a Heat formation - https://www.youtube.com/watch?v=PJweE3F_mPY
6.4b Heat reaction - https://www.youtube.com/watch?v=61fKkCwWu4s&list=PLp52f0A-lsC6yMHsGLejBroZ_6FUqJUNl&index=1

6.7 heating curve of water: Q1-Q5. --
https://www.youtube.com/watch?v=NflMkAkJUWI&list=PLp52f0A-lsC6yMHsGLejBroZ_6FUqJUNl&index=2

Lab - heat of reaction –
Part 1 – calibrate cup – later tater
Part 2 - heat of reaction of Mg + HCl -- https://www.youtube.com/watch?v=pjFV-A7wVJI
Biofuels - https://www.youtube.com/watch?v=OpEB6hCpIGM

Ch 7 The electron, periodicity.

7.1-7.3 Light, freq, wavelength, energy –
Light - problems $c = \lambda\nu$- https://www.youtube.com/watch?v=goJl54Y1hio
Quantized Energy $E = h\nu$ : https://www.youtube.com/watch?v=VLc20TF9jZw
Microwaves - https://www.youtube.com/watch?v=kp33ZprO0Ck
7.4 Bohr model – https://www.youtube.com/watch?v=au2HCVn9IJI
7.5 -7.11 Quantum Mechanics, Quantum numbers, electron configuration
A - https://www.youtube.com/watch?v=Aoi4j8es4gQ
B - https://www.youtube.com/watch?v=diCGRJskeDA
Schrodinger equation - https://www.youtube.com/watch?v=O6g-7rUgrdg
Schrodinger's cat - https://www.youtube.com/watch?v=UjaAxUO6-Uw&t=61s
7.12 Periodic Trends:
a - https://www.youtube.com/watch?v=hePb00CqvP0
b - https://www.youtube.com/watch?v=XK-WTYncldA
c -ionic radius - https://www.youtube.com/watch?v=lzKJWwTDU34
d ☺ https://www.youtube.com/watch?v=qc02d3G_wb4

7.12 – Speed of light - https://www.youtube.com/watch?v=pTn6Ewhb27k
7.13 – PES diagrams – AP test - https://www.youtube.com/watch?v=gyKD6QULa0A
7.13a – another PES -

Ch 8 Bonding, Lewis structures, molecular shapes, VSEPR
8.1 – Ionic bonding (lattice energy) - https://www.youtube.com/watch?v=hiyTfhjeF_U
8-10
Lewis Structures -single bonds: https://www.youtube.com/watch?v=qCOkHYYc4mE
Lewis Structures – double/triple bonds : https://www.youtube.com/watch?v=zEQeVvdjkBc
8-9
Sigma-pi https://www.youtube.com/watch?v=nTujP4jCbsg
VSEPR
a - https://www.youtube.com/watch?v=Q9-JjyAEqnU&t=290s
b https://www.youtube.com/watch?v=nxebQZUVvTg&t=645s
C. - https://www.youtube.com/watch?v=Moj85zwdULg&t=2s
D. https://www.youtube.com/watch?v=umwN74a0A2g'
E. https://www.youtube.com/watch?v=nLhmgF81Kqo
Snowflake bonding - https://www.youtube.com/watch?v=ao2Jfm35XeE
8.9 – water droplet - https://www.youtube.com/watch?v=ynk4vJa-VaQ
8.8 Bond Energy
BOND ENERGIES - https://www.youtube.com/watch?v=jr11SKQxBgc
BOND ENERGIES ex 1 - https://www.youtube.com/watch?v=eL6lS7uw7rc
BOND ENERGIES ex 2 - https://www.youtube.com/watch?v=HxWQOOBHcus

Ch 11 Colligative Properties – Solutions

11.5a Colligative Props 1
https://www.youtube.com/watch?v=hOhgXRukRDg
11.5b Colligative Props 2
https://www.youtube.com/watch?v=c8dDLe37ONg
11.5c Colligative Props 3
https://www.youtube.com/watch?v=wzAToPL8Src

Ch 12 – Rate of Reaction –

12.1 – Intro to Rate Laws…. - https://www.youtube.com/watch?v=WDXzVI8SmfE
12.1a – Rate Constant (k) - https://www.youtube.com/watch?v=eOSRn0jPbTk
12.1b – Integrated Rate Laws - https://www.youtube.com/watch?v=wYqQCojggyM
12.6a – Elementary Reactions - https://www.youtube.com/watch?v=0g1YwEbwifw
12.6b – Multistep Reactions - https://www.youtube.com/watch?v=I64dUjEzrcw
12.6c – Rate Limiting Step - https://www.youtube.com/watch?v=MEEg7aHqk6A
12.6d – Reaction Intermediate - https://www.youtube.com/watch?v=tzpStGaLjSE
12.7a – Activation Energy - https://www.youtube.com/watch?v=YacsIU97OFc
12.7b – Lab Ea – The Reaction Pathway – https://www.youtube.com/watch?v=0LjOXV4b-nk
12.8a – Catalyst classes – https://www.youtube.com/watch?v=OfP5h1T2KJI
12.8b – Catalysts - https://www.youtube.com/watch?v=KYD5LNVWne8
12.9 – Derivation of Integrated Rate Laws - https://www.youtube.com/watch?v=BXSa-el0CkA
12.10 - Beer's Law for rate law lab - https://www.youtube.com/watch?v=zuUvQN8KXOk

Ch 13 – Equilibrium

13.1 – Equilibrium - https://www.youtube.com/watch?v=cHAjhM3y3ds
13.2 – Equilibrium Constant - https://www.youtube.com/watch?v=xfGlEXWDRZE
13.3 – Equilibrium Disturbances - https://www.youtube.com/watch?v=dd5p0VZ-MZg
13.7 – Le Chatelier's Principle - https://www.youtube.com/watch?v=PciV_Wuh9V8
13.7a – ditto -- https://www.youtube.com/watch?v=XmgRRmxS3is
Haber Process - https://www.youtube.com/watch?v=EvknN89JoWo

Acid Base

14.1 – Acid base pH pOH intro - https://www.youtube.com/watch?v=NNXvokAcSuE
14.2 – pH pKa - https://www.youtube.com/watch?v=2zhvaGNJiJ8
14.2a – another - https://www.youtube.com/watch?v=kJTCuRSeh6g
14.2b – yet another - https://www.youtube.com/watch?v=XAmMGNPYVZM

Ch 15 Ksp
15.1 – Ksp first - https://www.youtube.com/watch?v=WjiXbemBXkE
15.2 – Common Ion - https://www.youtube.com/watch?v=qawipem0LwA
15.3 – Selective Precipitation - https://www.youtube.com/watch?v=6kAAXXHc4w8
15.4 – Problems PPT - https://www.youtube.com/watch?v=TBtaeyKkykU
Lab – Ksp lead(II) iodide - https://www.youtube.com/watch?v=icZFGVyUqXQ

Ch 16 Free Energy Entropy Thermodynamics G=H-TS

16.1  Entropy - https://www.youtube.com/watch?v=MALZTPsHSoo
16.1a – Spontaneity - https://www.youtube.com/watch?v=hNSD0YDsPsE
16.2. Gibbs free energy - https://www.youtube.com/watch?v=huKBuShAa1w
16.3 – Gibbs again - https://www.youtube.com/watch?v=8N1BxHgsoOw
16.3a – Gibbs problems - https://www.youtube.com/watch?v=fICs5vjBuX0
16.4 – Thermo $0^{th}$ Law - https://www.youtube.com/watch?v=S2hsaTO0bO8
16.5 – Thermo 3 Laws - https://www.youtube.com/watch?v=O7HwhkYt6YU
16.6 – Thermo problems worked - https://www.youtube.com/watch?v=FmYVHN6lHdg
16.7 - Entropy explained… https://www.youtube.com/watch?v=DxL2HoqLbyA
16.2.
16.3.

Electrochem

Lots problems - https://www.youtube.com/watch?v=_O1EKjTi8pA

17.1 – Intro electrochem - https://www.youtube.com/watch?v=2VT5rl8P84M
17.1a – dewitt intro elect - https://www.youtube.com/watch?v=teTkvUtW4SA

17.2 – aaa
17.3 – bbb

https://www.youtube.com/watch?v=tUJDbj9Vp5w
neil Armstrong lander

neil 2
https://www.youtube.com/watch?v=dNlZXso0-I4

Colligative Props 1
https://www.youtube.com/watch?v=hOhgXRukRDg

Colligative Props 2
https://www.youtube.com/watch?v=c8dDLe37ONg

Colligative Props 3
https://www.youtube.com/watch?v=wzAToPL8Src

alloys _ AP Chem
https://www.youtube.com/watch?v=PaGJwOPg2kU

physics – fluids drag
https://www.youtube.com/watch?v=GMmNKUlXXDs

In p2p1 video 1
https://www.youtube.com/watch?v=5HzCXAfKA8c

In p2p1 video 2
https://www.youtube.com/watch?v=xSHDawWK30s